

HW2 Controllable text-to-music generation

R13942143 何景盛

Code: https://drive.google.com/drive/folders/1MbOW7QDElQyHRSV_j0OqA7qp65-nWERT?usp=drive_link

Outline

- **Target Music**
 - Evaluation (Meta Audiobox Aesthetics)
- **Retrieval**
 - Methods
 - Retrieval Results
 - Evaluation (CLAP, Meta Audiobox Aesthetics, Melody similarity)
- **Generation**
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 - Modification
 - Evaluation (CLAP, Meta Audiobox Aesthetics, Melody similarity)
- **Results Analysis**

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Evaluation (Meta Audiobox Aesthetics)

Target music	CE	CU	PC	PQ
【菊花台-周杰倫（D調）】 ...	7.5835	7.9325	4.7412	7.8742
10_country_114_beat_4-4.wav	6.1745	7.6489	2.827	7.3412
4_jazz_120_beat_3-4.wav	5.6374	7.6761	3.069	7.5009
Hedwig's theme x dizi ...	7.2268	7.5771	3.4659	7.5737
Mussorgsky : Pictures at ...	7.3856	7.9087	2.8347	7.7107
Spirited Away OST 「Always ...	7.8315	8.1262	3.2204	7.9907
6_rock_102_beat_3-4.wav	6.8696	8.041	3.2327	7.7582
竹笛 这世界那么多人 ...	7.6358	8.0111	3.9627	7.7681
【楽譜あり】IRIS OUT ...	7.6261	7.9778	3.858	7.9531

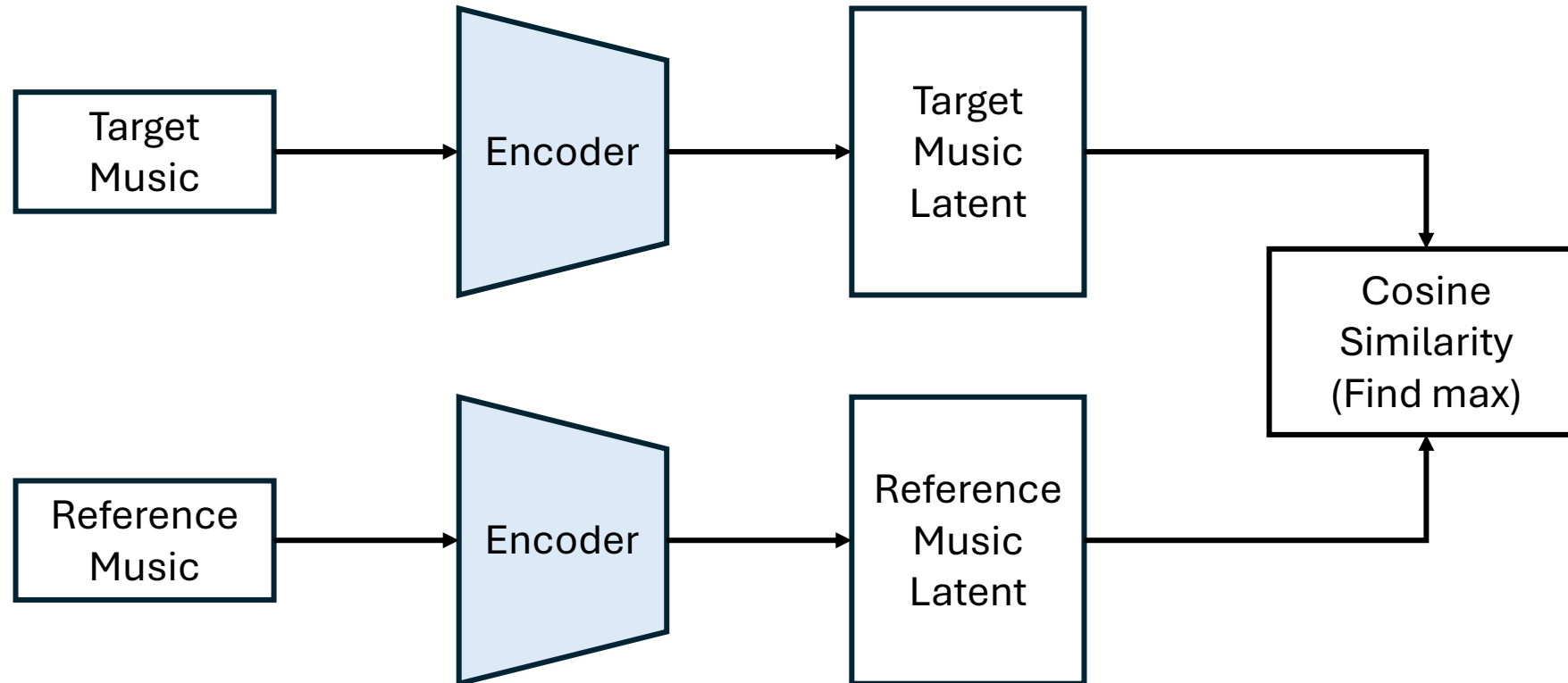
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Methods

1. The retrieval process begins by encoding both the target music and the reference music into a latent space using **pre-trained encoders**.
 - **Music2Latent**: The audio is transformed into a 64-dimensional latent representation, segmented into multiple temporal parts. The mean across all segments is taken to form the final 64-dimensional latent vector.
 - **CLAP**: Each is also encoded into a 512-dimensional latent embedding.
2. The **Cosine Similarity** is employed to measure the closeness between the target and reference embeddings. The reference embedding with the highest similarity score to each target is selected as the most relevant match.
3. Evaluation is conducted between the reference music and the target music using the following metrics:
 - **CLAP**
 - reference music and target music
 - **Meta Audiobox Aesthetics**
 - CE (Content Enjoyment)
 - CU (Content Usefulness)
 - PC (Production Complexity)
 - PQ (Production Quality)
 - **Melody similarity (accuracy)**

Methods (Diagram)



Retrieval Results

Target music	Reference (Music2latent)	Reference (CLAP)
【菊花台-周杰倫（D調）】 ...	MIDI-Unprocessed_SMF_02_R1_2004_01-05_ORIG_MID--AUDIO_02_R1_2004_05_Track05_wav.wav	燕子.mp3
10_country_114_beat_4-4.wav	6_jazz-funk_116_fill_4-4.wav	146_latin-brazilian-baiao_95_beat_4-4.wav
4_jazz_120_beat_3-4.wav	185_reggae_78_fill_4-4.wav	79_latin-reggaeton_90_beat_4-4.wav
Hedwig's theme x dizi ...	鄉歌.mp3	漁舟唱晚 (笛).mp3
Mussorgsky : Pictures at ...	MIDI-Unprocessed_XP_21_R1_2004_03_ORIG_MID--AUDIO_21_R1_2004_04_Track04_wav.wav	MIDI-Unprocessed_XP_21_R1_2004_03_ORIG_MID--AUDIO_21_R1_2004_04_Track04_wav.wav
Spirited Away OST 「Always ...	MIDI-Unprocessed_SMF_02_R1_2004_01-05_ORIG_MID--AUDIO_02_R1_2004_08_Track08_wav.wav	MIDI-Unprocessed_XP_21_R1_2004_03_ORIG_MID--AUDIO_21_R1_2004_04_Track04_wav.wav
6_rock_102_beat_3-4.wav	3_funk_80_beat_4-4.wav	82_neworleans-funk_84_beat_4-4.wav
竹笛 这世界那么多人 ...	MIDI-Unprocessed_SMF_05_R1_2004_01_ORIG_MID--AUDIO_05_R1_2004_03_Track03_wav.wav	元若藍-愛X無限大-03.綠柚子.mp3
【楽譜あり】IRIS OUT ...	MIDI-Unprocessed_SMF_05_R1_2004_01_ORIG_MID--AUDIO_05_R1_2004_02_Track02_wav.wav	MIDI-Unprocessed_XP_20_R2_2004_01_ORIG_MID--AUDIO_20_R1_2004_03_Track03_wav.wav

Evaluation (CLAP)

Target music	CLAP (Music2latent ref)	CLAP (CLAP ref)
【菊花台-周杰倫（D調）】 ...	0.6266	0.7116
10_country_114_beat_4-4.wav	0.6743	0.9111
4_jazz_120_beat_3-4.wav	0.6321	0.9101
Hedwig's theme x dizi ...	0.7157	0.4882
Mussorgsky : Pictures at ...	0.8629	0.8065
Spirited Away OST 「Always ...	0.6563	0.7482
6_rock_102_beat_3-4.wav	0.9266	0.9586
竹笛 这世界那么多人 ...	0.4109	0.4578
【楽譜あり】IRIS OUT ...	0.5122	0.7162

Evaluation (Meta Audiobox Aesthetics-CLAP)

Target music	CE (ref)	CU (ref)	PC (ref)	PQ (ref)
【菊花台-周杰倫（D調）】 ...	7.0114	7.7988	4.4588	8.0251
10_country_114_beat_4-4.wav	6.7984	8.0512	3.3687	7.8177
4_jazz_120_beat_3-4.wav	6.4788	8.0021	2.996	7.7039
Hedwig's theme x dizi ...	7.0704	7.8155	3.5833	7.9515
Mussorgsky : Pictures at ...	7.3371	7.781	2.7809	7.5532
Spirited Away OST 「Always ...	7.3371	7.781	2.7809	7.5532
6_rock_102_beat_3-4.wav	6.6629	8.0368	3.3677	7.8015
竹笛 这世界那么多人 ...	7.6094	7.8588	6.2029	8.2581
【楽譜あり】IRIS OUT ...	7.2053	7.6396	3.3015	7.2844

Evaluation (Meta Audiobox Aesthetics-Music2latent)

Target music	CE (ref)	CU (ref)	PC (ref)	PQ (ref)
【菊花台-周杰倫（D調）】 ...	7.4269	7.6292	3.315	7.3612
10_country_114_beat_4-4.wav	5.3996	7.548	2.5248	7.8141
4_jazz_120_beat_3-4.wav	4.1299	7.1218	1.9045	7.5694
Hedwig's theme x dizi ...	7.302	6.7932	4.0454	6.9894
Mussorgsky : Pictures at ...	7.3371	7.781	2.7809	7.5532
Spirited Away OST 「Always ...	7.4415	7.7526	3.492	7.4893
6_rock_102_beat_3-4.wav	7.134	8.1295	3.2141	7.939
竹笛 这世界那么多人 ...	6.1829	7.0639	3.5594	7.0246
【樂譜あり】IRIS OUT ...	7.2897	7.5536	3.4317	7.4072

Evaluation (Melody similarity)

Target music	Melody similarity (Music2latent ref)	Melody similarity (CLAP ref)
【菊花台-周杰倫（D調）】 ...	0.125	0.1418
10_country_114_beat_4-4.wav	0.1641	0.099
4_jazz_120_beat_3-4.wav	0.121	0.0715
Hedwig's theme x dizi ...	0.1435	0.0965
Mussorgsky : Pictures at ...	0.1696	0.1696
Spirited Away OST 「Always ...	0.1709	0.1137
6_rock_102_beat_3-4.wav	0.1082	0.1
竹笛 这世界那么多人 ...	0.1061	0.0223
【楽譜あり】IRIS OUT ...	0.1717	0.0383

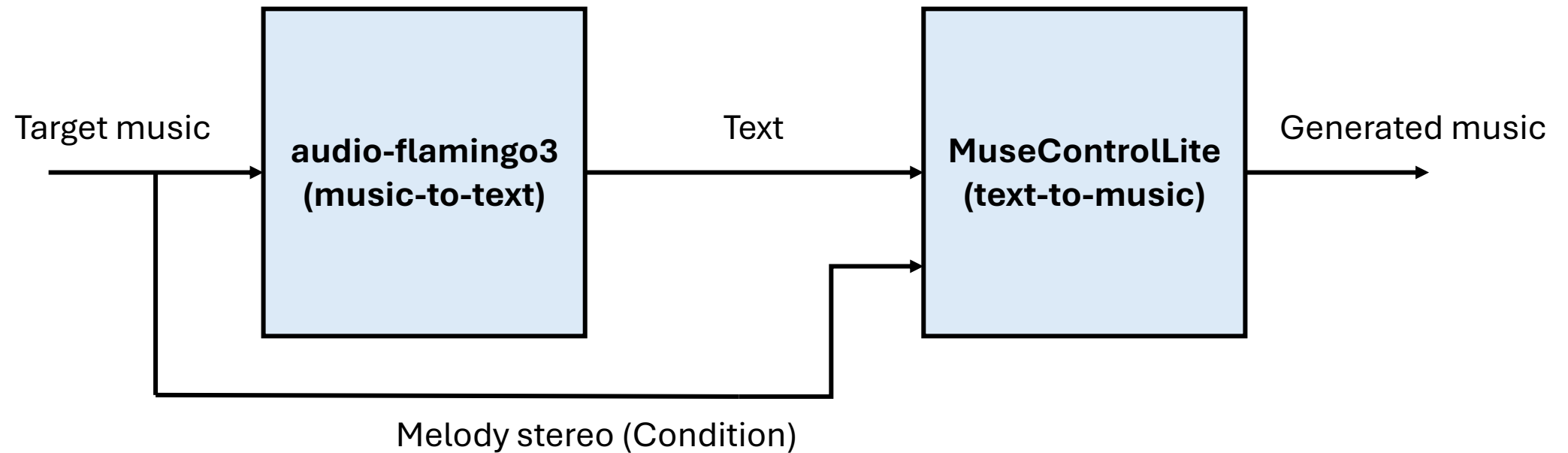
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Methods

1. The generation process begins with a music-to-text conversion using **audio-flamingo3** in think mode. The target music is transformed into a detailed music caption.
2. The text-to-music model **MuseControlLite** is employed to generate music from the obtained textual description and melody condition.
3. The generated music is then evaluated against the target music using multiple metrics, as follows:
 - **CLAP**
 - Target music and text
 - Text and generated music
 - Generated music and target music (Since MuseControlLite generates 47s audio samples, each target music is trimmed to 47s.)
 - **Meta Audiobox Aesthetics**
 - CE (Content Enjoyment)
 - CU (Content Usefulness)
 - PC (Production Complexity)
 - PQ (Production Quality)
 - **Melody similarity (accuracy)**

Methods (Diagram)



Modification

- Due to limited GPU (RTX2080ti) memory capacity, the audio-flamingo3 model was quantized to 4-bit precision in order to reduce VRAM consumption and enable efficient inference.
- The figure on the right shows the code added in the implementation.

```
66     from transformers import BitsAndBytesConfig
67     quant_config = BitsAndBytesConfig(
68         load_in_4bit=True,
69         bnb_4bit_use_double_quant=True,
70         bnb_4bit_quant_type="nf4",
71         bnb_4bit_compute_dtype=torch.float16,
72     )
73     max_mem = {
74         "0": "10GiB",
75         "cpu": "48GiB"
76     }
77     model = llava.load(
78         model_path,
79         device_map="auto",
80         torch_dtype=torch.float16,
81         quantization_config=quant_config,
82         max_memory=max_mem
83     )
```

Path: audio-flamingo/llava/cli/infer_audio.py

Evaluation (CLAP)

Target music	Target and text	Text and generated	Generated and target
【菊花台-周杰倫（D調）】 ...	0.472	0.5224	0.6579
10_country_114_beat_4-4.wav	0.5779	0.5277	0.8306
4_jazz_120_beat_3-4.wav	0.4247	0.3938	0.7644
Hedwig's theme x dizi ...	0.5915	0.5899	0.7461
Mussorgsky : Pictures at ...	0.3906	0.5812	0.6391
Spirited Away OST 「Always ...	0.481	0.5947	0.7411
6_rock_102_beat_3-4.wav	0.4814	0.469	0.8652
竹笛 这世界那么多人 ...	0.4504	0.5628	0.7785
【楽譜あり】 IRIS OUT ...	0.4684	0.5399	0.6566

Evaluation (Meta Audiobox Aesthetics)

Target music	CE (gen)	CU (gen)	PC (gen)	PQ (gen)
【菊花台-周杰倫（D調）】 ...	7.3843	7.9407	3.4921	7.524
10_country_114_beat_4-4.wav	6.807	7.995	3.9637	7.5715
4_jazz_120_beat_3-4.wav	6.1558	7.6527	4.6441	7.4016
Hedwig's theme x dizi ...	6.8433	7.4098	2.5816	6.7791
Mussorgsky : Pictures at ...	7.486	8.0452	2.5908	7.978
Spirited Away OST 「Always ...	7.6733	8.1133	3.0904	8.1144
6_rock_102_beat_3-4.wav	6.7009	8.0229	4.0113	7.6011
竹笛 这世界那么多人 ...	7.5174	7.9961	3.5639	7.9505
【楽譜あり】IRIS OUT ...	7.3203	8.1547	4.3961	8.139

Evaluation (Melody similarity)

Target music	Melody similarity (gen)
【菊花台-周杰倫（D調）】 ...	0.7682
10_country_114_beat_4-4.wav	0.3363
4_jazz_120_beat_3-4.wav	0.6568
Hedwig's theme x dizi ...	0.8788
Mussorgsky : Pictures at ...	0.8324
Spirited Away OST 「Always ...	0.6556
6_rock_102_beat_3-4.wav	0.2444
竹笛 这世界那么多人 ...	0.8533
【楽譜あり】 IRIS OUT ...	0.6899

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Results Analysis (CLAP)

- **Analysis**

- The retrieval method using CLAP achieved the highest performance (0.745).
- The generation method also reached a comparable CLAP score (0.736), indicating nearly equivalent semantic alignment with the target music.
- This demonstrates that the generated music preserves a high level of semantic consistency with the target.

- **Conclusion**

- Both methods exhibit strong semantic consistency with the target music.

Methods	Mean CLAP
Retrieval (CLAP)	0.745
Retrieval (Music2latent)	0.664
Generation	0.736

Results Analysis (Meta Audiobox Aesthetics)

- **Analysis**

- Generation methods achieve aesthetic scores comparable to the target music, demonstrating stable quality.

- **Conclusion**

- Music generated by MuseControlLite attains aesthetic scores on par with real music.

Music	Mean CE	Mean CU	Mean PC	Mean PQ
Target	6.42	7.88	3.15	7.72
Retrieval (CLAP)	7.06	6.97	3.34	7.77
Generation	7.1	7.92	3.59	7.67

Results Analysis (Melody similarity)

- **Analysis**

- The retrieval method primarily captures semantic features, resulting in substantial differences in melodic structure.
- In contrast, the generation method, conditioned on the target melody, achieves a significantly higher melody similarity score.

- **Conclusion**

- The generation approach (with melody conditioning) greatly outperforms retrieval in preserving melodic structure.
- This demonstrates the model's ability to generate music that faithfully aligns with the target melody.

Methods	Mean Melody similarity
Retrieval (Music2latent ref)	0.136
Retrieval (CLAP ref)	0.088
Generation (melody-conditioned)	0.645

References

- <https://github.com/LAION-AI/CLAP>
- <https://github.com/facebookresearch/audiobox-aesthetics>
- https://github.com/NVIDIA/audio-flamingo/tree/audio_flamingo_3
- <https://github.com/fundwotsai2001/MuseControlLite>